

Financial Mathematics Formula Sheet

An independent reference for interest theory and Exam FM preparation

Standard notation: i = effective interest rate per period, $v = (1 + i)^{-1}$, $d = i/(1 + i)$, and n = number of periods.

1. Interest Measurement

If $a(t)$ is the accumulation function with $a(0) = 1$, then an investment of K at time 0 has value $Ka(t)$ at time t .

$$i_{[t_1, t_2]} = \frac{a(t_2) - a(t_1)}{a(t_1)}, \quad d_{[t_1, t_2]} = \frac{a(t_2) - a(t_1)}{a(t_2)}.$$

For one period,

$$v = \frac{1}{1 + i} = 1 - d, \quad d = \frac{i}{1 + i}, \quad i = \frac{d}{1 - d}.$$

Simple interest:

$$a(t) = 1 + it.$$

Simple discount:

$$a(t) = \frac{1}{1 - dt}.$$

Compound interest:

$$a(t) = (1 + i)^t, \quad v^t = (1 + i)^{-t}.$$

For nominal rates convertible m times per year,

$$1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m, \quad 1 - d = \left(1 - \frac{d^{(m)}}{m}\right)^m.$$

For a constant force of interest δ ,

$$1 + i = e^\delta, \quad \delta = \ln(1 + i).$$

For a time-varying force δ_t ,

$$\delta_t = \frac{a'(t)}{a(t)}, \quad a(t) = \exp\left(\int_0^t \delta_s ds\right).$$

2. Equations of Value

Move every cash flow to one common comparison date. If cash flow C_k occurs at time t_k , then at time 0,

$$PV = \sum_k C_k v^{t_k}.$$

At time T ,

$$FV_T = \sum_k C_k (1 + i)^{T - t_k}.$$

The yield rate or internal rate of return is a rate i satisfying

$$\sum_k C_k (1 + i)^{-t_k} = 0,$$

where inflows and outflows are assigned opposite signs.

3. Level Annuities

For payments of 1 at times $1, 2, \dots, n$,

$$a_{\overline{n}|} = \frac{1 - v^n}{i}, \quad s_{\overline{n}|} = \frac{(1 + i)^n - 1}{i}.$$

For payments of 1 at times $0, 1, \dots, n - 1$,

$$\ddot{a}_{\overline{n}|} = (1 + i)a_{\overline{n}|} = \frac{1 - v^n}{d},$$

$$\ddot{s}_{\overline{n}|} = (1 + i)s_{\overline{n}|}.$$

Perpetuities:

$$a_{\infty|} = \frac{1}{i}, \quad \ddot{a}_{\infty|} = \frac{1}{d}.$$

A deferred annuity-immediate with first payment after k deferral periods has present value

$${}_k|a_{\overline{n}|} = v^k a_{\overline{n}|}.$$

For level payment R , multiply the appropriate annuity factor by R .

4. Varying Annuities

For payments $1, 2, \dots, n$ at times $1, 2, \dots, n$,

$$(Ia)_{\overline{n}|} = \frac{\ddot{a}_{\overline{n}|} - nv^n}{i}.$$

For payments $n, n - 1, \dots, 1$,

$$(Da)_{\overline{n}|} = \frac{n - a_{\overline{n}|}}{i}.$$

For payments $1, (1 + g), \dots, (1 + g)^{n-1}$,

$$PV = \frac{1 - \left(\frac{1+g}{1+i}\right)^n}{i - g}, \quad i \neq g.$$

For an infinite geometric payment stream beginning with 1 at time 1,

$$PV = \frac{1}{i - g}, \quad i > g.$$

5. Payment Frequency

If the annual effective rate is i , define

$$i^{(m)} = m((1 + i)^{1/m} - 1),$$

$$d^{(m)} = m(1 - (1 + i)^{-1/m}).$$

For m payments per year of amount $1/m$ over n years,

$$a_{\overline{n}|}^{(m)} = \frac{1 - v^n}{i^{(m)}} = a_{\overline{n}|} \frac{i}{i^{(m)}},$$

$$\ddot{a}_{\overline{n}|}^{(m)} = \frac{1 - v^n}{d^{(m)}} = a_{\overline{n}|} \frac{i}{d^{(m)}}.$$

For a continuous payment rate $c(t)$,

$$PV = \int_0^n c(t)v(t) dt.$$

If the force is constant, $v(t) = e^{-\delta t}$.

6. Loans

For a loan of amount L repaid by n level payments R ,

$$L = Ra_{\overline{n}|}, \quad R = \frac{L}{a_{\overline{n}|}}.$$

Immediately after payment t , the outstanding balance may be found prospectively:

$$B_t = Ra_{\overline{n-t}|},$$

or retrospectively:

$$B_t = L(1+i)^t - Rs_{\overline{t}|}.$$

For nonlevel payments,

$$B_t = B_{t-1}(1+i) - R_t.$$

Interest and principal in payment t are

$$I_t = iB_{t-1}, \quad P_t = R_t - I_t.$$

Principal repaid from time r through time s is the decrease in balance over that interval.

7. Bonds

Let F be face value, r the coupon rate per coupon period, C the redemption value, j the yield rate per coupon period, and n the number of remaining coupons. Coupon amount is Fr .

Bond price:

$$P = Fr a_{\overline{n}|} \big|_{i=j} + Cv_j^n.$$

Equivalent premium/discount form:

$$P - C = (Fr - Cj)a_{\overline{n}|} \big|_{i=j}.$$

Thus $P > C$ when $Fr > Cj$ (premium) and $P < C$ when $Fr < Cj$ (discount).

Book value immediately after coupon t :

$$B_t = Fr a_{\overline{n-t}|} \big|_{i=j} + Cv_j^{n-t}.$$

Recursive book value:

$$B_t = B_{t-1}(1+j) - Fr.$$

Interest earned during period t is jB_{t-1} ; the difference between the coupon and interest is the premium amortization or discount accumulation.

8. Spot and Forward Rates

If s_n is the n -period spot rate,

$$v(0, n) = (1 + s_n)^{-n}.$$

If $f_{n,n+m}$ is the m -period forward rate beginning at time n ,

$$(1 + s_{n+m})^{n+m} = (1 + s_n)^n (1 + f_{n,n+m})^m.$$

For one-period forward rates,

$$(1 + s_n)^n = \prod_{k=1}^n (1 + f_{k-1,k}).$$

Value any deterministic cash flow stream using spot rates:

$$P = \sum_t \frac{C_t}{(1 + s_t)^t}.$$

9. Duration and Convexity

For price

$$P(i) = \sum_t C_t(1+i)^{-t},$$

Macaulay duration is

$$D_M = \frac{\sum_t t C_t(1+i)^{-t}}{P}.$$

Modified duration is

$$D_{\text{mod}} = \frac{D_M}{1+i} = -\frac{1}{P} \frac{dP}{di}.$$

A useful convexity measure is

$$C = \frac{1}{P} \frac{d^2 P}{di^2} = \frac{\sum_t t(t+1)C_t(1+i)^{-t-2}}{P}.$$

For a small change Δi ,

$$\frac{\Delta P}{P} \approx -D_{\text{mod}} \Delta i + \frac{1}{2} C (\Delta i)^2.$$

10. Matching and Immunization

Exact matching: choose asset cash flows so that the asset cash flow at each relevant date equals the liability cash flow at that date.

First-order immunization at rate i :

- present value of assets = present value of liabilities;
- modified duration of assets = modified duration of liabilities.

A standard Redington condition additionally requires asset convexity to exceed liability convexity at the matching rate.

11. Quick Identities

$$\ddot{a}_{\overline{n}|} - a_{\overline{n}|} = 1 - v^n, \quad \ddot{s}_{\overline{n}|} - s_{\overline{n}|} = (1+i)^n - 1.$$

$$a_{\overline{n}|} = v\ddot{a}_{\overline{n}|}, \quad s_{\overline{n}|} = (1+i)^n a_{\overline{n}|}.$$

For a level annuity, always check whether the valuation date is *before the first payment* (immediate) or *at the first payment* (due).